universität innsbruck
Department of Computer Science

Introduction

Data Engineering and Analytics

Eva Zangerle

Context

Data analysis is a process of inspecting, cleansing, transforming, and modelling data with the goal of discovering useful information, informing conclusions, and supporting decision-making.[1]

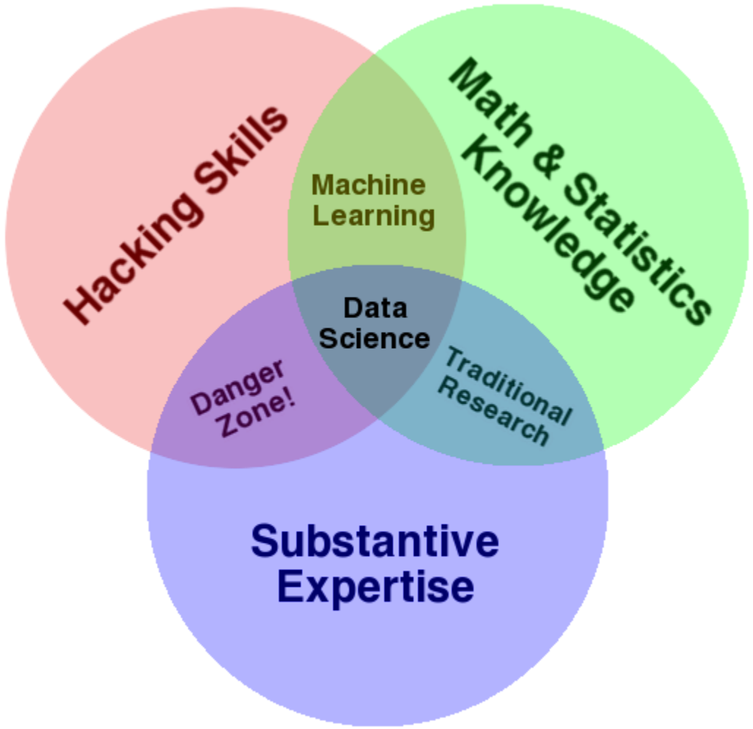
Source: https://en.wikipedia.org/wiki/Data_analysis

Context cont.

Data science is an interdisciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data,[1][2] and apply knowledge and actionable insights from data across a broad range of application domains. Data science is related to data mining, machine learning and big data.

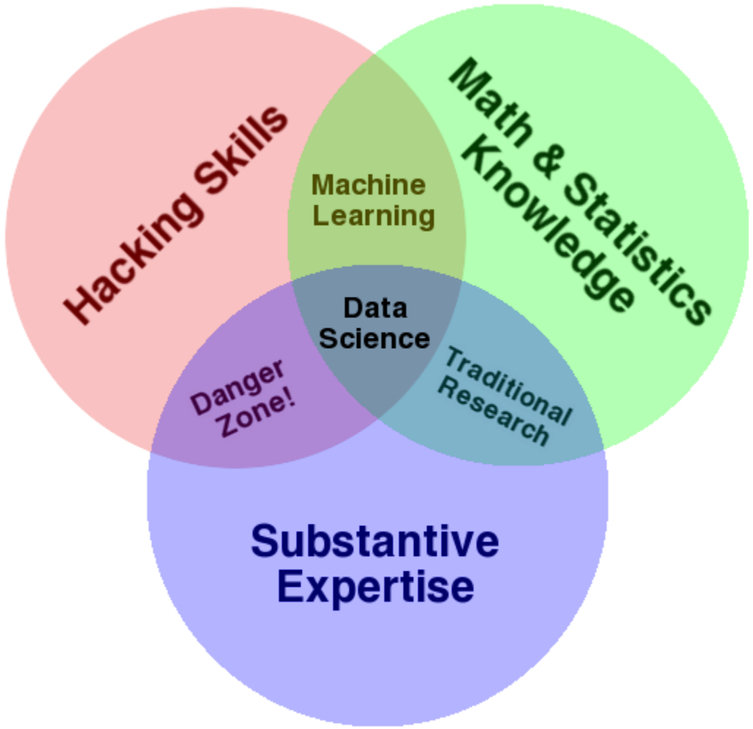
Source: https://en.wikipedia.org/wiki/Data_science

Data Science Venn Diagram



Source: http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagramhttps://towardsdatascience.com/why-the-data-science-venn-diagram-is-misleading-16751f852063

Data Science Venn Diagram



Also, soft skills are very important in data science, as they enable effective communication, collaboration, and problem-solving within teams and with stakeholders.

> Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise.” (John Tukey)

Source: http://drewconway.com/zia/2013/3/26/the-data-science-venn-diagramhttps://towardsdatascience.com/why-the-data-science-venn-diagram-is-misleading-16751f852063

Data Engineering and Analytics

We need to not only apply methods, but also

- look at results,
- interprete,
- reflect, and
- derive actionable steps from our observations.

Asking Questions

Problem Statement

- Most important step of the data science pipeline.
- Defines which questions should be asked and answered.
- Sets the goal of the analyses.
- Defines the scope of the analysis.
- Identifies the stakeholders involved in the analysis.
- Determines the data requirements for the analysis.
- Determines the appropriate methods and tools for the analysis.
- Plans the data collection and preparation steps.
- ....

Problems are often vaguely defined

- Understand the problem that needs to be solved.
- Assess situation with respect to the problem.
  - Requirements?
  - Assumptions? Constraints?
  - Available resources (personnel, data, computing power)?
- Understand risks and benefits of problem.
  - Main costs?
  - Potential benefits?
  - Risks?
- Define success criteria to assess the solution.
  - When is the project done?
  - What is the ultimate goal?
  - What can actually be measured?

Analysis Workflow

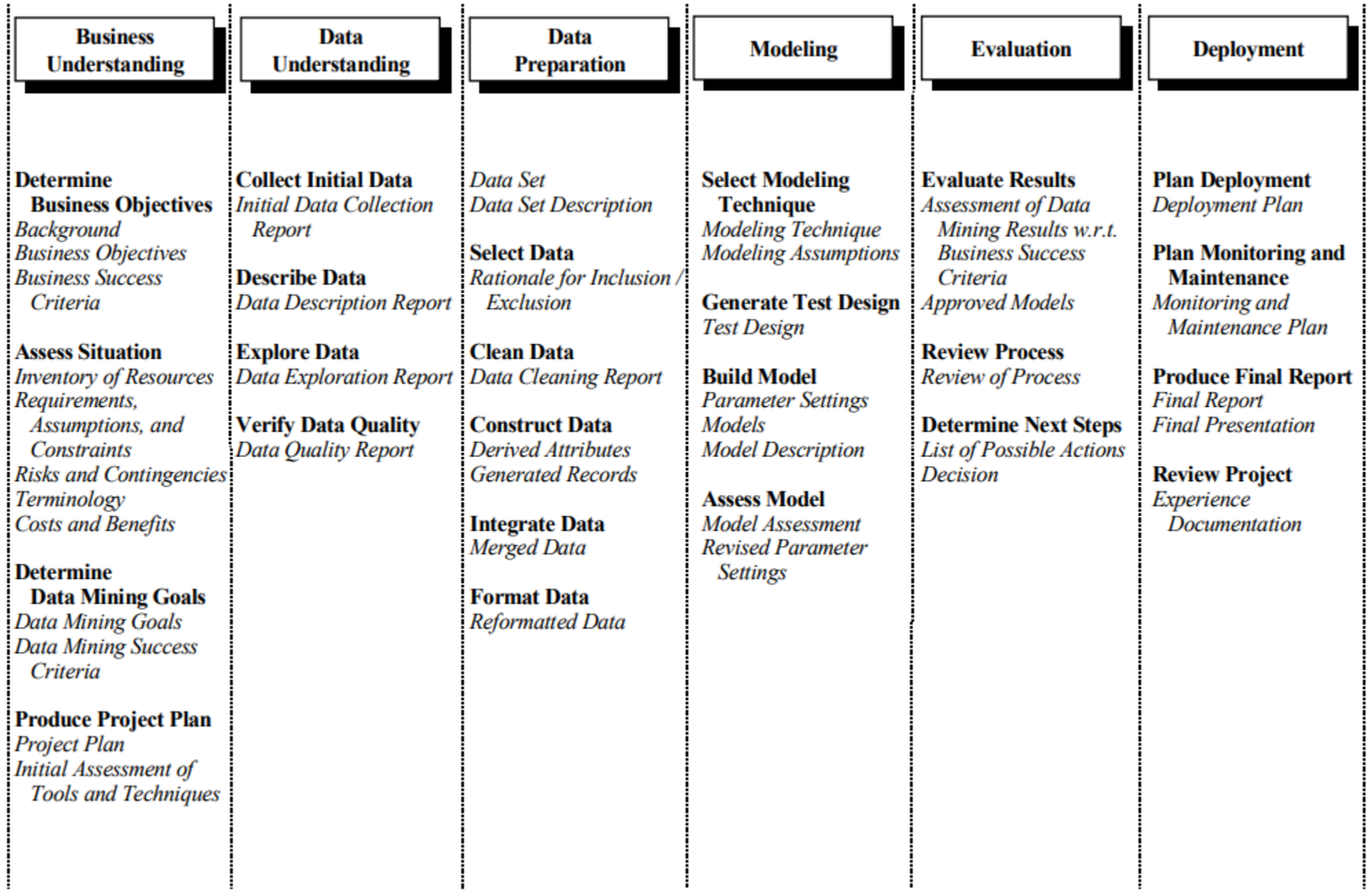
Process Models

- Describe widely used approach for data analytics (like a template approach).
- Cross-industry standard process for data mining (CRISP-DM)



"CRISP DM Process Diagram" by Kenneth Jensen is licensed under CC SA 3.0

CRISP-DM



Wirth, Rüdiger, and Jochen Hipp. "CRISP-DM: Towards a standard process model for data mining." Proceedings of the 4th international conference on the practical applications of knowledge discovery and data mining. Vol. 1. London, UK: Springer-Verlag, 2000.

Process Models

Extension of CRISP-DM, emphasizes the back-and-forth at any stage.

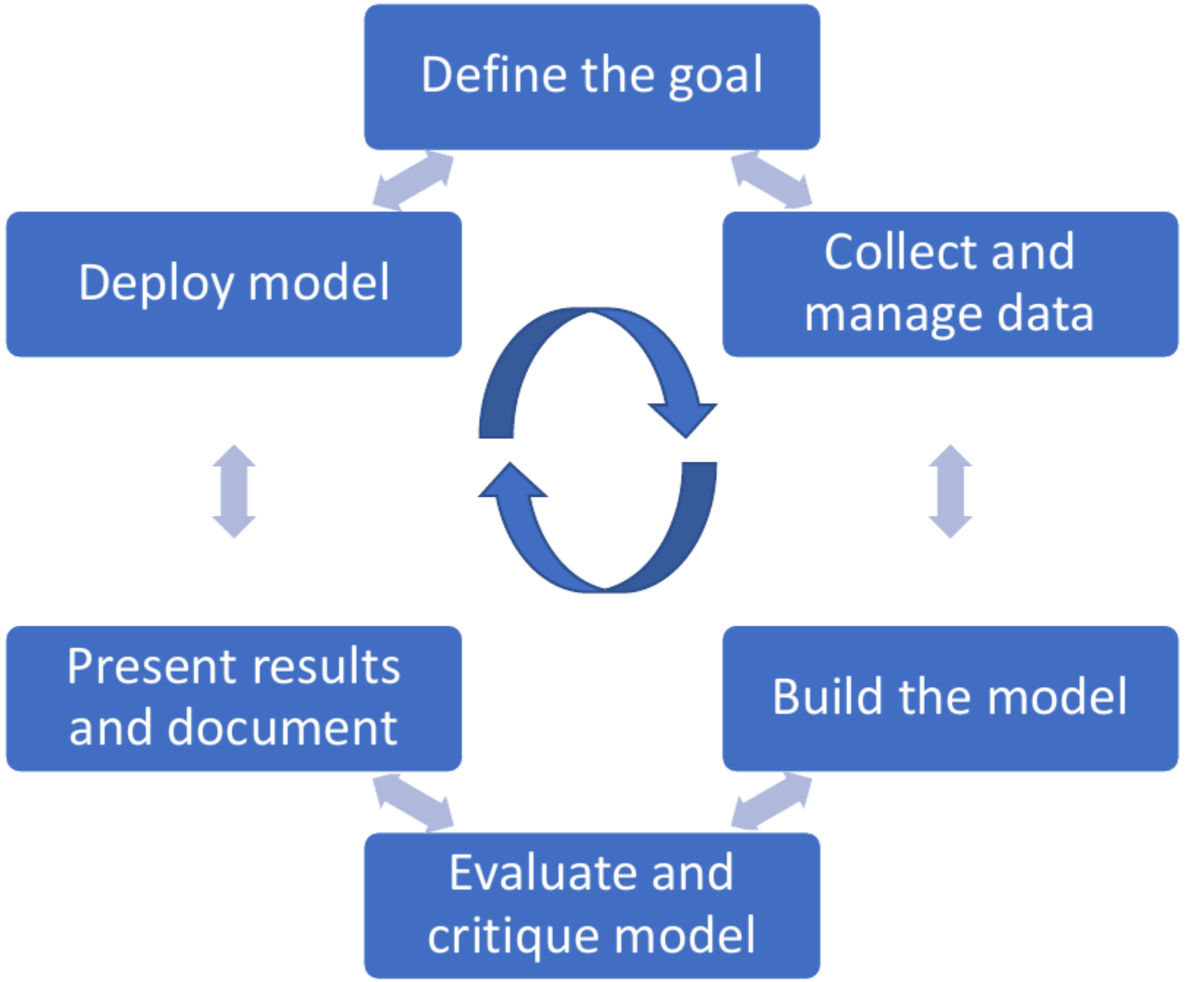


Figure based on Practical Data Science with R; Nina Zumel, John Mount; Manning Publications; 2nd Edition, 2017.

Process Models

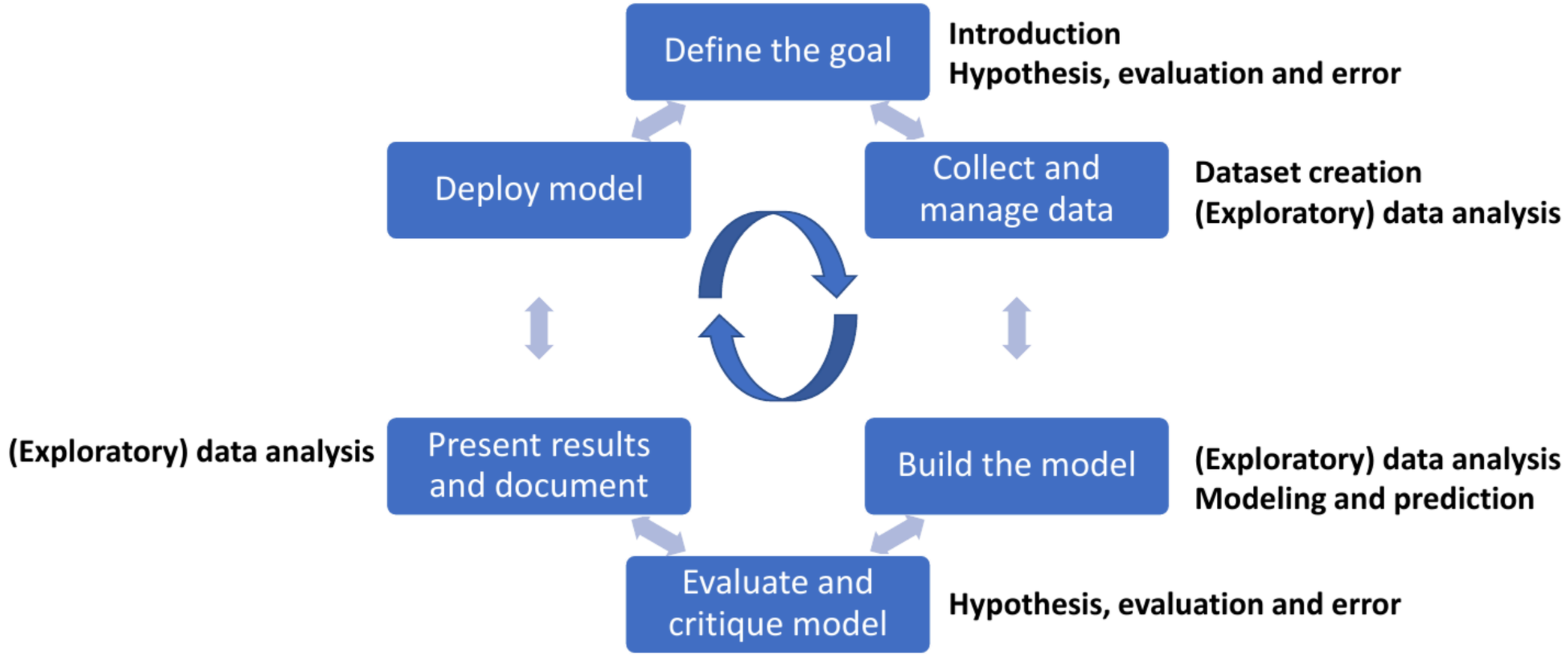


Figure based on Practical Data Science with R; Nina Zumel, John Mount; Manning Publications; 2nd Edition, 2017.

Data Analysis Workflow

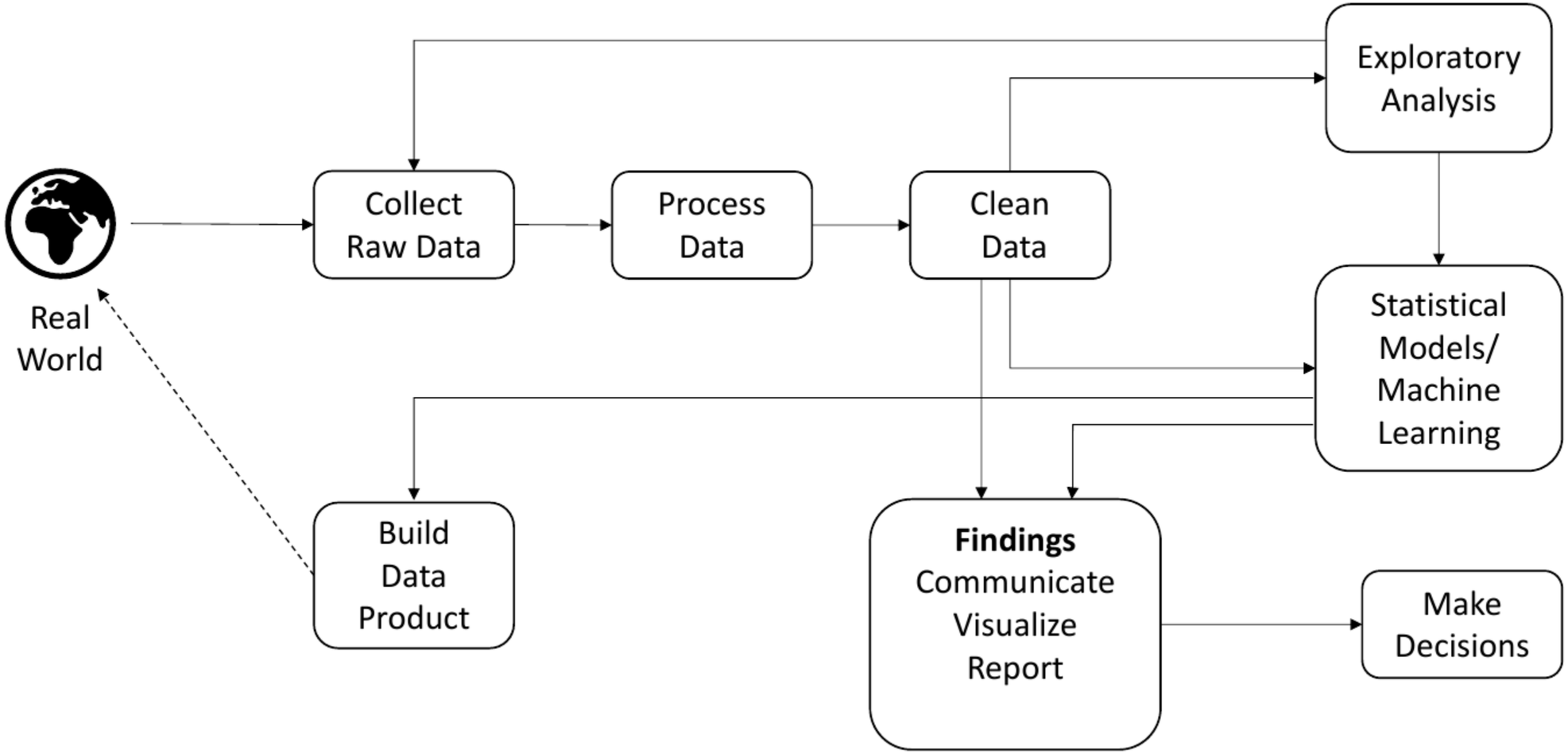


Figure adapted from Doing Data Science, Cathy O'Neil, Rachel Schutt, O'Reilly, 2013.

Data Analysis

- First step usually exploratory data analysis (EDA) to
  - Detect mistakes
  - Check assumptions
  - Do a preliminary selection of suitable models
  - Determine relationships among explanatory variables
  - Assess direction and rough size of relationships between explanatory and outcome variables.
- Exploratory data analysis precedes formal (confirmatory) data analysis.

Data Analysis

- Formal (confimatory) analyses
  - Researcher has hypothesis (specific idea) of how exploratory and outcome variables relate
  - Based on models → ideal, mathematical representations of observable characteristics
  - Models have two components:
    - Structural component: Relationship between explanatory variables and outcome variables
    - Error component: captures deviations of individual observations from mean
  - Model captures how distribution of outcome varies with changes in explanatory variables.

Data Analysis

- Models make assumptions about population where sample comes from.
- → critical to test

Confirmatory Data Analysis

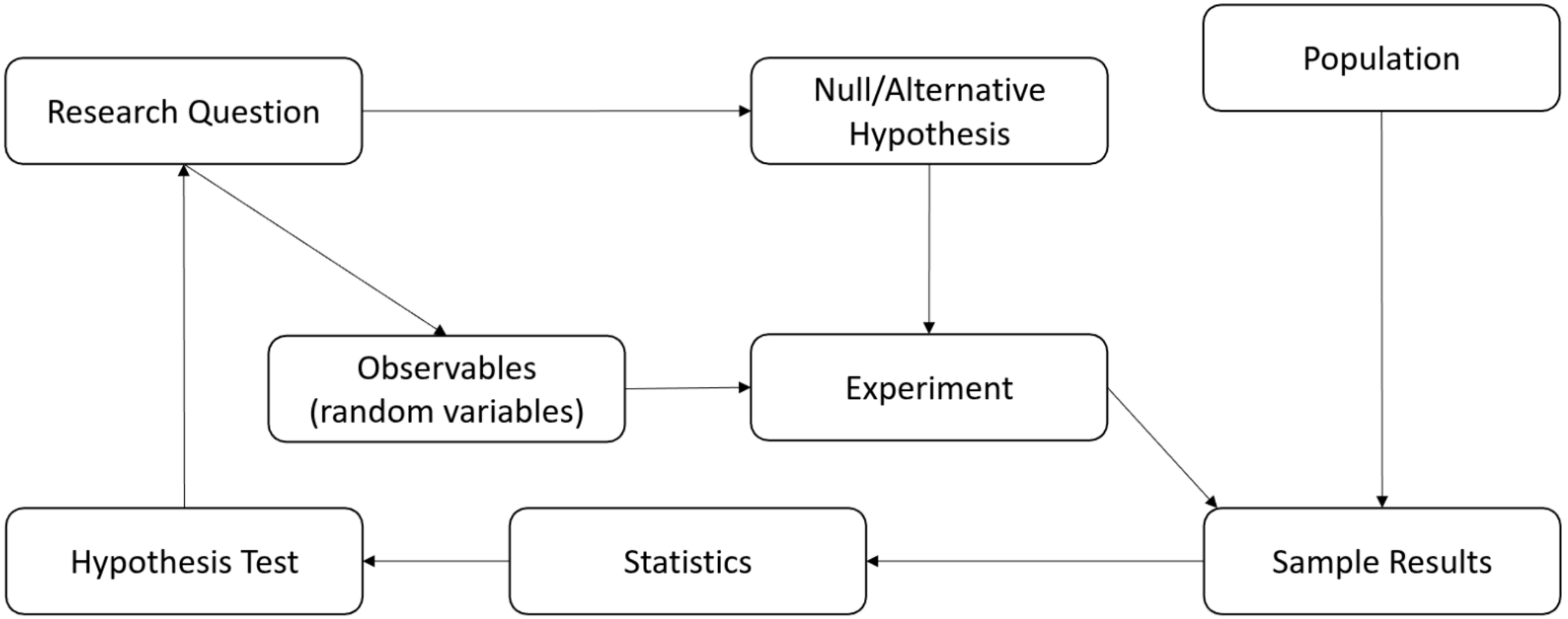


Figure adapted from Experimental Design and Analysis, Howard J. Seltman, 2018.

Jupyter Notebooks

- Available at https://git.uibk.ac.at/informatik/dbis/dbis-teaching/data-engineering-analytics-notebooks
- Short demo
- See `README.md`

Further Reading

- How to ask the right questions as a data scientist (link)
- The Data Science Venn Diagram (link)
- Why the Data Science Venn Diagram is Misleading (link)